

Medical Readmission Data Dashboard

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Course:

Data Storytelling For Varied Audiences - D601

Date:

March, 2026

Dataset and Project Scenario

- Dataset includes information on
10,000 hospital patients
- Variables include demographics,
medical conditions, hospital services,
and readmission status
- Readmission is defined as a patient
returning to the hospital within 30
days after discharge
- Goal: identify patterns and factors
that contribute to hospital
readmissions

Dashboard Overview

- Interactive Tableau dashboard created to analyze readmission trends

- Dashboard highlights two key performance indicators (KPIs):

Readmission Rate - percentage of patients who return within 30 days.

Total Readmissions - total number of patients readmitted within 30 days.

- Visualizations compare readmission rates across admission type, age group, diabetes status, and complication risk.
- Map visualization identifies geographic patterns in readmissions
- Filters allow users to explore the data for specific patient groups

Readmission Rate by Admission Type

- Compares readmission rates across emergency, elective, and observation admissions
- Helps identify operational patterns that may affect readmission risk
- Emergency admissions often involve more complex or urgent medical conditions

Why this visualization was chosen:

- Bar chart format allows easy comparison between categories
- Enables decision-makers to quickly identify which admission types have higher readmission rates

Readmission Rate by State

- Map visualization shows geographic differences in readmission rates
- Helps hospital leaders identify regional patterns or potential problem areas
- Supports decisions related to resource allocation and patient support programs

Why this visualization was chosen:

- Maps provide a clear geographic perspective
- Makes it easier to identify patterns across multiple hospital locations

Interactive Filters

The dashboard includes filters that allow users to explore the data interactively.

Filter 1: Admission Type

- Allows users to focus on specific admission categories
- Updates all charts to display results for the selected admission type

Filter 2: Diabetes Status

- Allows analysis of readmission patterns among patients with and without diabetes
- Helps identify how chronic conditions may impact readmission rates

Dashboard Accessibility and Design Choices

- Dashboard uses a colorblind-friendly color palette to ensure accessibility for users with color vision deficiencies
- Clear chart titles and labels help viewers quickly understand each visualization
- Consistent layout guides the viewer through the information in a logical order
- KPIs are positioned at the top to provide a quick overview of performance

Reflection and Storytelling

Two elements of effective storytelling
used in this presentation:

- Clarity – Simple visualizations help viewers quickly understand key insights
- Narrative structure – Dashboard begins with overall metrics and then explores contributing factors

Lessons Learned

- Designing dashboards requires balancing data accuracy with clear communication
- Presenting data effectively requires explaining both insights and design decisions

References

Tableau Software. (2023). Tableau Desktop Documentation.

Professional Communication

This document was written to be clear, organized, and aligned to the D601 Task 2 rubric requirements